Causal selected sub-goals extraction from the offline learning world model

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***Abstract*—Transfer learning has become an effective solution for the sparse reward problem in reinforcement learning by enabling agents to leverage previous knowledge for new tasks through goal-conditional reinforcement learning. However, the issue of Out-of-Distribution (OOD) has led to the need for learning robust and essential features. While techniques such as representation learning have proven effective, the learned features lack semantic explanations, making them challenging to interpret and utilize. To address this issue, This paper proposes a novel feature extraction algorithm that utilizes a trained world model from a counterfactual perspective to extract a set of understandable state-action pairs as sub-goals. The states of these sub-goals are then clustered in the latent space, and an MLP is used to establish a mapping between states and actions, which guides the agent's exploration in a new environment.**

***Index Terms*—Sub-goal discovery, Reinforcement learning, Counterfactual.**

I. Introduction

The out-of-distribution (OOD) problem has become an important concern because it violates the $i.i.d$ assumptions on source and target data, resulting in reduced generalization of the learned knowledge [1]. Therefore, learning robust representations is necessary to deal with OOD problems and to achieve better performance in downstream tasks with distributional changes in the test dataset, while at the same time, difficult-to-understand representations are one of the important challenges for in-depth and further robotic research [2].

II. Related Works

Causal representation learning. Machine learning algorithms are often limited to capturing correlations between variables rather than causal relationships. Capturing causal and essential representations plays an important role in domain generalization.

Sub-goal discovery. To address sparse rewards, the sub-goals have been introduced to enable agents with intrinsic reward signals, under the goal-conditional RL framework.

III. Motivation

The lack of intuitive interpretability of self-supervised causal representational learning is a significant challenge to delving into the reliability of representational learning. Inspired by the human strategy of splitting into subtasks based on key steps when dealing with complex tasks, we might extract state-action pairs as sub-goals based on causal counterfactual principle in an easily understandable state space, as shown in Figure 1, i.e., capable of reflecting agents' strategic decisions, which also provides researchers with unique insights.

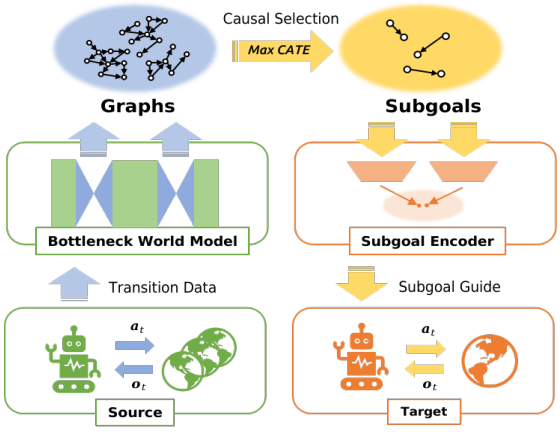


Fig. 1. Diagram of causal selected sub-goals extraction algorithm.

IV. Method

Intuitively, in the transition dynamic graph model, there are some behaviors ((s, a) pairs) that are necessary for completing the task across all the different semantic-similar environments. Therefore, we attempt to extract sub-goals in the graphical model from the counterfactual perspective as the representation of this task. To be specific, from the causal lens, those behaviors have causal effects on the final accumulate reward.

A. Sub-goals extraction algorithm.

World model. To construct the world model in the latent space, we adopt a pre-trained variational auto-encoder and a pre-trained PPO agent to generate expert demonstrations across k different environments E^k in Figure 2. After collecting all the trajectories $\mathcal{D} = \{\{\tau_i^e\}_{i=1}^{N_e} | e \in E^k\}$, where each trajectory consists of a sequence of environment-specific state-action pairs $\tau^e = (z_t^e, a_t)_{t=0,\dots}$, we feed the data from each environment trajectory into the GRU-MDN framework. The goal is to minimize the log-likelihood of the weighted Gaussian mixture model between the output of the MDN $\hat{z}_{t+1}$ and the ground truth z_{t+1}. This can be achieved by minimizing the following loss function $\mathcal{L}_{wm} = \mathbb{E}[\log \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(z_i|\mu_k, \sigma_k)]$, where K, μ and σ represent the number and parameters of the Gaussian mixture models, respectively, and π is the coefficient parameter of the Gaussian mixture model.

a) Discrete graphical world model. As the latent variable z and the output of GRU are continuous vectors, we introduce the quantized bottleneck network which is a variant of auto-

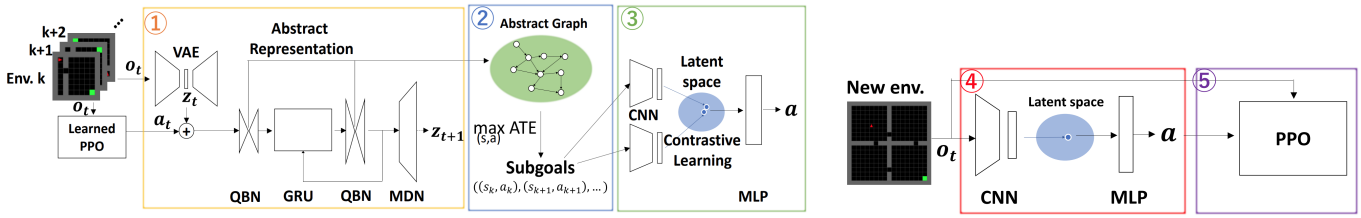


Fig. 2. **Left**: the extraction process involves three main blocks. First, we learn a bottleneck world model from expert demonstrations. Second, we extract sub-goals that maximize the conditional average treatment effect (CATE) using a latent transmit dynamic graph. Third, we cluster the sub-goals by contrastive learning. **Right**: guide a new PPO agent learning using the pre-trained CNN encoder.

encoder, $Q(x) = D(quantized(E(x))$. In the final layer of the QBN encoder, the activate function is $\phi(x') = 1.5tanh(x') + 0.5tanh(-3x')$, which quantized the continuous variables into d bits and 3-level representation. QBN trained by the L_2 norm $\mathcal{L}_{qbn} = ||x - Q(x)||^2$ to obtain discrete latent representation.

After the pre-training warm-up, we insert the ObsQBN before and the HxQBN after the GRU in our world model. Then, we input the trajectories $\{\tau^e\}^{N_e}$ into the evaluation mode of the world model. By using the latent vector of the ObsQBN as an edge and the latent vector of the HxQBN as a node, we can extract an abstract the Moore machine graphical model $\mathcal{G}^e(V, E)$, where V represents the hidden state s^e of the GRU, and E represents the state-action (z^e, a) pairs.

b) Select the causal sub-goal. Mimicking the expert's every step behavior is unnecessary, according to the causal counterfactual mechanism, capturing across-environmental important behaviors is sufficient, so-called causal sub-goals.

Definition 1. (Causal sub-goals) *a set of state-action pairs g which has a specific causal effect ($> \epsilon$) on final accumulated reward value across all the training environment E^k.*

$$\delta^e = \mathbb{E}[R|do((s^e, a) \in \mathcal{G}^e)] - \mathbb{E}[R|do((s^e, a) \notin \mathcal{G}^e)])\} \tag{1}$$

$$g^e = \{(s^e, a)|\delta^e > \epsilon)\} \tag{2}$$

$$g = \{g^e|e \in E^k\} \tag{3}$$

Here, R is the final reward function, and ϵ is the threshold of a causal effect. We define the expectation formulation as follows: $\mathbb{E}[R|do((s,a) \in \mathcal{G}] = \frac{1}{N}\sum_{i=1}^{N} R(s_i^{end}) \cdot f_{(s,a)}(s_i^{end})$ and $\mathbb{E}[R|do((s,a) \notin \mathcal{G}] = \frac{1}{N}\sum_{i=1}^{N} R(s_i^{end}) \cdot g_{(s,a)}(s_i^{end})$, where s_i^{end} is the end node of trajectory τ_i, and N is the number of end nodes in $\mathcal{G}$. The reward of end state s_i is denoted by $R(s_i^{end})$. $f_{(s,a)}(s_i^{end})$ and $g_{(s,a)}(s_i^{end})$ are the target reachable possibility of the agent in the original $\mathcal{G}$ and the (s,a) pair-removed $\mathcal{G}$, respectively. The reachable possibility is computed by a DFS (Depth First Search) algorithm.

Definition 2. (Target reachable possibility) *If $\exists\hat{\tau}(s,a)$ from the start node and end node in $\mathcal{G}^e$, the reachable possibility is equal to 1, otherwise 0.*

Empirically, the redundant nodes and edges are inevitably produced during this process, we delete incorrect and redundant connections in $\mathcal{G}^e$ by comparing the transition dynamic between dataset and latent space. Choose the node-edge pair whose CATE value is larger than ϵ as a sub-goal in g^e.

B. Contrastive learning and transfer learning.

After we extract the sub-goals set g, and then reconstruct the observation set O_g from g. In order to learn a sub-goals representation encoder ϕ_{cnn}, we cluster the similar observation in the latent space by contrastive learning objective: $\mathcal{L}_{con} = \frac{1}{2N}\sum_{i=1}^{N}\sum_{j=1}^{N} y_{i,j}d_{i,j}^2 + (1 - y_{i,j})[\max(m - d_{i,j}, 0)]^2$. The binary category function $y_{i,j}$ follows $\mathcal{L}_1$ loss between two observations, distance function $d_{i,j}$ follows $\mathcal{L}_2$ loss.

Next, train ψ_{mlp} to reconstruct action from g. So far, we got the sub-goals knowledge encoder $f_{\phi_{cnn},\psi_{mlp}} : \mathcal{O} \to \mathcal{A}$.

To transfer f to an agent in a new environment task, we introduce auxiliary cross-entropy loss between policy f and π: $\mathcal{L}_{aux} = -\sum_j f \cdot \log(\pi_{ppo}(a|s))$, where j represent the number of action types. While training policy π, updates loss function $\mathcal{L}^p = \mathcal{L}_{ppo} + \beta_p\mathcal{L}_{aux}$ with the training step p, ($\beta_{p+1} = \eta\beta_p, \beta_0 > 0, \eta < 1$) to trade-off exploration and exploitation.

V. Result

The result shows that the extracted sub-goals are open-door and across-door behaviors. And the sample efficiency apparently improved, shown in Fig 3. During this experiment, we observe the result of extraction isn't always stable, it might be because some redundant connections still exist in $\mathcal{G}^e$.

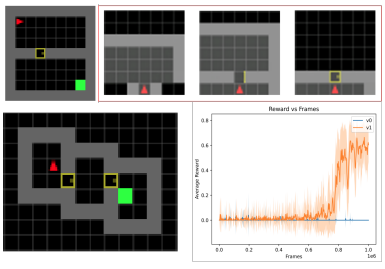


Fig. 3. **Upper** left: training environment; right: extracted sub-goals; **Down** left: new environment; right: reward value. (blue line: vanilla PPO, orange line: PPO with sub-goals knowledge.)

Acknowledgements

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References

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